Quantum Boltzmann Machines

From Quantum Neural Networks to Quantum Generative Models

Morten Hjorth-Jensen

Department of Physics & CCSE
University of Oslo

May 20, 2026

Outline

From Discriminative to Generative Models

Discriminative QNN (what we had)

- Input x encoded via $U(x)$
- Output: class label or regression value
- Minimise: $\mathcal{L} = \sum_k (f(x_k) - y_k)^2$
- Goal: learn $p(y|x)$

Generative QML (our goal now)

- **No** input encoding needed
- Output: full probability distribution $p(x)$
- Minimise: $D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, p_\theta)$
- Goal: learn to *sample* from p_{data}

Generative models in QML:

- Quantum Boltzmann Machines (QBM) — *this talk*
- Quantum Circuit Born Machines (QCBM)
- Quantum GANs
- Quantum Variational Autoencoders

Key difference:
QBMs model distributions via a *thermal state* (density matrix) rather than a pure variational state.

Classical Restricted Boltzmann Machine

An RBM has **visible** units $\mathbf{v} \in \{0,1\}^n$ and **hidden** units $\mathbf{h} \in \{0,1\}^m$.

Energy function:

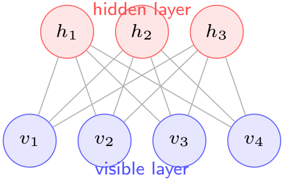
$$E(\mathbf{v},\mathbf{h}) = -\sum_i a_i v_i - \sum_j b_j h_j - \sum_{ij} v_i W_{ij} h_j$$

Joint distribution and partition function:

$$p(\mathbf{v},\mathbf{h}) = \frac{e^{-\beta E}}{Z}, \quad Z = \sum_{\mathbf{v},\mathbf{h}} e^{-\beta E}$$

Marginal over visible:

$$p(\mathbf{v}) = \frac{1}{Z} \sum_{\mathbf{h}} e^{-\beta E(\mathbf{v},\mathbf{h})}$$



No intra-layer connections $\Rightarrow$ tractable conditional independence:

$$p(h_j = 1|\mathbf{v}) = \sigma\big(b_j + \sum_i W_{ij} v_i\big)$$

RBM Training: Contrastive Divergence

Log-likelihood gradient:

$$\frac{\partial \ln p(\mathbf{v})}{\partial W_{ij}} = \langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_{\text{model}}$$

- Data term: easy (one forward pass)
- Model term: hard (requires summing over all $\mathbf{v}, \mathbf{h}$)

Contrastive Divergence (CD-k):

1. Clamp $\mathbf{v}^{(0)} = \mathbf{v}_{\text{data}}$
2. Run k steps of Gibbs sampling $\mathbf{v}^{(0)} \to \mathbf{h}^{(0)} \to \mathbf{v}^{(1)} \to \cdots \to \mathbf{v}^{(k)}$
3. Approximate model term with $\mathbf{v}^{(k)}$

CD-k update rule

$$\Delta W_{ij} \propto \langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_k$$

Similarly for biases a_i, b_j.

Key limitation

The partition function Z is intractable for large systems — gradient estimates are biased. Quantum hardware promises an exact, unbiased estimate of $\langle v_i h_j \rangle_{\text{model}}$ via *quantum sampling*.

From Classical to Quantum Gibbs States

Replace binary units with **qubits** and the classical energy function with a **quantum Hamiltonian**:

Classical: $p(\mathbf{v}) = e^{-\beta E(\mathbf{v})}/Z$

Quantum: The thermal (Gibbs) state

$$\boxed{\rho(\theta) = \frac{e^{-\beta H(\theta)}}{\mathrm{Tr}\left[e^{-\beta H(\theta)}\right]}}$$

where $H(\theta)$ is a parametrised quantum Hamiltonian.

The partition function is now a *quantum trace*:

$$Z(\theta) = \mathrm{Tr}\left[e^{-\beta H(\theta)}\right] = \sum_k e^{-\beta \epsilon_k(\theta)}$$

where $\{\epsilon_k\}$ are eigenvalues of $H(\theta)$.

Degrees of freedom

Classical RBM	QBM
Binary spins $\{0,1\}^n$	n qubits
Energy $E(\mathbf{v},\mathbf{h})$	Hamiltonian $H(\theta)$
Gibbs dist. $e^{-\beta E}/Z$	Gibbs state $e^{-\beta H}/Z$
Marginal $p(\mathbf{v})$	Reduced state ρ_v

Key gain

Quantum off-diagonal (transverse) terms in $H(\theta)$ allow the QBM to represent *quantum correlations* and *superpositions* — beyond any classical BM.

The QBM Hamiltonian

The most general parametrised n-qubit QBM Hamiltonian:

$$H(\theta) = -\sum_i \theta_i^x X_i - \sum_i \theta_i^z Z_i - \sum_{i<j} \theta_{ij}^{zz} Z_i Z_j - \cdots$$

- θ_i^z: **longitudinal fields** (diagonal) — classical bias terms, analogous to a_i
- θ_{ij}^{zz}: **ZZ couplings** (diagonal) — classical two-body interactions, analogous to W_{ij}
- θ_i^x: **transverse fields** (off-diagonal) — purely quantum term, no classical analogue

Thermal state in the eigenbasis

$$H(\theta) |\epsilon_k\rangle = \epsilon_k |\epsilon_k\rangle$$

$$\rho(\theta) = \sum_k \frac{e^{-\beta\epsilon_k}}{Z} |\epsilon_k\rangle \langle\epsilon_k|$$

Classical limit

Setting $\theta_i^x = 0$ for all i recovers a classical BM: $[H, Z_i] = 0 \; \forall i$, so ρ is diagonal in the Z-basis and reduces to a classical Boltzmann distribution.

Visible, Hidden, and Auxiliary Qubits

Following the classical RBM structure, the n qubits are partitioned into:

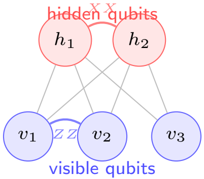
$$\mathcal{H} = \mathcal{H}_v \otimes \mathcal{H}_h, \quad n_v + n_h = n$$

- **Visible** qubits v: coupled to the data; we observe their reduced state $\rho_v = \mathrm{Tr}_h[\rho]$
- **Hidden** qubits h: latent variables that are traced out

The visible marginal is the model distribution:

$$p_\theta(\mathbf{v}) = \langle \mathbf{v} | \, \rho_v(\theta) \, | \mathbf{v} \rangle$$

where $\rho_v(\theta) = \mathrm{Tr}_h\left[\frac{e^{-\beta H(\theta)}}{Z}\right]$.



Quantum connections can include XX, YY, ZZ within visible, within hidden, and across layers.

The QBM Training Objective

Given a dataset with empirical distribution $\pi(\mathbf{v})$, we want $p_\theta(\mathbf{v}) \approx \pi(\mathbf{v})$.
Minimise the KL divergence:

$$\mathcal{L}(\theta) = D_{\mathrm{KL}}(\pi \,\|\, p_\theta) = \sum_{\mathbf{v}} \pi(\mathbf{v}) \ln \frac{\pi(\mathbf{v})}{p_\theta(\mathbf{v})}$$

Expanding using $p_\theta(\mathbf{v}) = \langle \mathbf{v}| \rho_v |\mathbf{v}\rangle$:

$$\mathcal{L}(\theta) = -\sum_{\mathbf{v}} \pi(\mathbf{v}) \ln p_\theta(\mathbf{v}) + \text{const}$$

$$= -\mathrm{Tr}[\sigma_{\text{data}} \ln \rho_v(\theta)] + \text{const}$$

where $\sigma_{\text{data}} = \sum_{\mathbf{v}} \pi(\mathbf{v}) |\mathbf{v}\rangle \langle \mathbf{v}|$.

Quantum relative entropy

The training objective is the **quantum relative entropy**:

$$\mathcal{L} = S(\sigma_{\text{data}} \,\|\, \rho_v(\theta))$$

$$= \mathrm{Tr}[\sigma_{\text{data}}(\ln \sigma_{\text{data}} - \ln \rho_v)]$$

This reduces to the classical KL divergence when both states are diagonal.

Full QBM objective

Use the *total* quantum relative entropy:

$$\mathcal{L} = S(\rho_{\text{target}} \,\|\, \rho(\theta))$$

when the target is itself a quantum state

Gradient of the Log-Likelihood

The gradient of the negative log-likelihood w.r.t. parameter θ_μ is:

$$\frac{\partial \mathcal{L}}{\partial \theta_\mu} = \langle \Lambda_\mu \rangle_{\rho_{\text{clamp}}} - \langle \Lambda_\mu \rangle_{\rho(\theta)}$$

where Λ_μ is the **generalised force** (operator conjugate to θ_μ) defined by:

$$H(\theta) = \sum_\mu \theta_\mu \Lambda_\mu$$

and the two expectation values are:

$$\langle \Lambda_\mu \rangle_{\text{clamp}} = \text{Tr}[\Lambda_\mu \, \rho_{\text{clamp}}] \quad \text{(data-dependent)}$$
$$\langle \Lambda_\mu \rangle_\theta = \text{Tr}[\Lambda_\mu \, \rho(\theta)] \quad \text{(model-dependent)}$$

Clamped state

ρ_{clamp} is the Gibbs state of the *same* Hamiltonian but with visible qubits clamped to a data sample:

$$\rho_{\text{clamp}} = \frac{e^{-\beta H^{(v)}}}{Z^{(v)}}$$

where $H^{(v)}$ is $H(\theta)$ with visible-qubit operators replaced by their classical values.

Classical analogy

This is the quantum analogue of:

$$\frac{\partial \ln p}{\partial W_{ij}} = \langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_{\text{model}}$$

The Quantum Log-Likelihood Gradient: Derivation

Key identity. For $H(\theta)$ depending smoothly on θ_μ:

$$\frac{\partial}{\partial\theta_\mu} \ln Z = -\beta \,\mathrm{Tr}[\Lambda_\mu \,\rho(\theta)] = -\beta\langle\Lambda_\mu\rangle_\theta$$

Using $p_\theta(\mathbf{v}) = \langle\mathbf{v}|\,\rho_v\,|\mathbf{v}\rangle$:

$$\frac{\partial}{\partial\theta_\mu} \sum_{\mathbf{v}} \pi(\mathbf{v}) \ln p_\theta(\mathbf{v}) = \sum_{\mathbf{v}} \pi(\mathbf{v}) \frac{\partial}{\partial\theta_\mu} \ln p_\theta(\mathbf{v})$$

After using the **Golden-Thompson inequality** and the **Duhamel formula** for matrix exponential derivatives:

$$\frac{\partial\mathcal{L}}{\partial\theta_\mu} = \beta\left[\langle\Lambda_\mu\rangle_{\mathrm{clamp}} - \langle\Lambda_\mu\rangle_\theta\right]$$

Physical interpretation

Each gradient step reduces the difference between the *clamped* (data-driven) and *free* (model) thermal averages of Λ_μ — the quantum generalisation of Contrastive Divergence.

Challenges: The Partition Function and Thermal Sampling

The intractability problem

Computing $\langle \Lambda_\mu \rangle_\theta = \mathrm{Tr}[\Lambda_\mu \rho(\theta)]$ requires:

$$\rho(\theta) = \frac{e^{-\beta H(\theta)}}{\mathrm{Tr}\left[e^{-\beta H(\theta)}\right]}$$

- Diagonalising an $2^n \times 2^n$ matrix — exponential cost classically
- Computing the matrix exponential and its trace
- Repeated for every gradient step

Three main approaches:

1. Quantum hardware sampling
2. Variational approximation of $\rho(\theta)$
3. Semiclassical / mean-field approximations

Quantum hardware advantage

A quantum device *naturally* prepares thermal states via cooling / environment coupling. Measuring $\mathrm{Tr}[\Lambda_\mu \rho]$ requires only $O(\varepsilon^{-2})$ shots — no diagonalisation.

Variational approach

Approximate the Gibbs state by a variational ansatz:

$$\rho(\theta) \approx \rho_\phi = U(\phi)\rho_0 U^\dagger(\phi)$$

Jointly optimise (θ, ϕ) — this is the **Quantum-Classical Boltzmann Machine**.

Method 1: Quantum imaginary-time evolution (QITE)
Evolve with the non-unitary operator $e^{-\Delta\tau H}$, approximated by a sequence of unitary gates using TDVP:

$$\rho(\tau) = \frac{e^{-\tau H}}{Z(\tau)}, \quad \tau = \beta/2$$

Each QITE step:

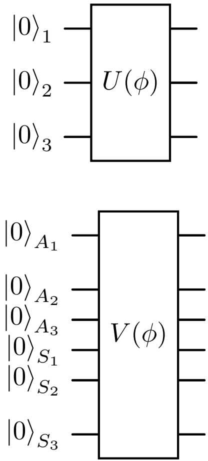
$$e^{-\Delta\tau H}\left|\psi\right\rangle \approx U_{\mathrm{QITE}}\left|\psi\right\rangle$$

where U_{QITE} is a local unitary found by solving the TDVP linear system $\sum_j F_{ij}\dot{\theta}_j = C_i$.

Method 2: Variational thermal ansatz

$$\rho_\phi = \mathrm{Tr}_A\left[U(\phi)\left|0\right\rangle_{SA}\left\langle 0\right| U^\dagger(\phi)\right]$$

Use ancilla qubits A to purify the mixed state ρ via a



Top: direct variational state. Bottom: thermofield double

Purification and the Thermofield Double State

Any mixed state ρ on a system S can be **purified**: there exists a pure state $|\Psi\rangle_{SA}$ on $S \otimes A$ such that:

$$\rho_S = \mathrm{Tr}_A[|\Psi\rangle\langle\Psi|]$$

For the Gibbs state, the canonical purification is the **thermofield double** (TFD):

$$|\mathrm{TFD}(\beta)\rangle = \frac{1}{\sqrt{Z(\beta)}} \sum_k e^{-\beta\epsilon_k/2} |\epsilon_k\rangle_S |\epsilon_k\rangle_A$$

Tracing out A:

$$\mathrm{Tr}_A[|\mathrm{TFD}\rangle\langle\mathrm{TFD}|] = \rho(\beta)$$

TFD as a QNN circuit

The TFD state can be prepared by a parametrised circuit on $2n$ qubits (n system $+\, n$ ancilla):

$$|\mathrm{TFD}(\beta)\rangle \approx V(\phi)\,|0\rangle^{\otimes 2n}$$

with $V(\phi)$ trained to minimise the free energy:

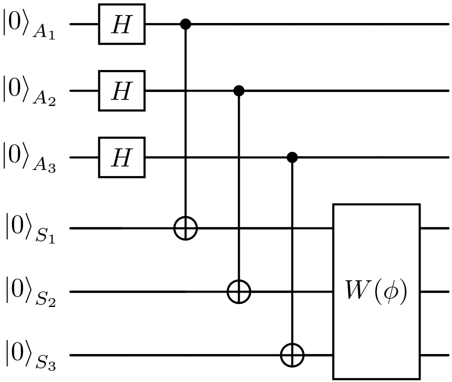
$$\mathcal{F}(\phi) = \langle H\rangle_\phi - \beta^{-1} S_{\mathrm{vN}}(\rho_\phi)$$

where $S_{\mathrm{vN}}(\rho) = -\mathrm{Tr}[\rho \ln \rho]$ is the von Neumann entropy.

Key connection

TFD Circuit: Quantikz Diagram

The thermofield double state prepared on $n = 3$ system qubits $+\ 3$ ancilla qubits:



- **Step 1:** $H^{\otimes n}$ on ancillas $\rightarrow$ uniform superposition
- **Step 2:** CNOT from each ancilla to its system partner $\rightarrow$ maximally entangled Bell pairs
- **Step 3:** Variational circuit $W(\phi)$ on system qubits tilts toward the correct thermal distribution

Variational QBM: Full Algorithm

The **Variational QBM** jointly optimises:

1. **Model parameters** θ: couplings in $H(\theta)$
2. **Thermal circuit parameters** ϕ: prepares $\rho(\beta, \theta) \approx \rho_\phi$

Algorithm (one step):

1. Sample $\mathbf{v}$ from dataset
2. Run TFD circuit $V(\phi)$ to get $\rho_\phi \approx \rho(\theta, \beta)$
3. Clamp visible qubits to $\mathbf{v}$; run $V_{\mathrm{clamp}}(\phi)$ to get ρ_{clamp}
4. Estimate $\langle \Lambda_\mu \rangle_\phi$ and $\langle \Lambda_\mu \rangle_{\mathrm{clamp}}$ by sampling
5. Update θ by gradient descent
6. Update ϕ by imaginary-time TDVP

Two nested optimisations

$$\min_\theta \mathcal{L}(\theta) = D_{\mathrm{KL}}(\pi \,\|\, p_\theta)$$

$$\min_\phi \mathcal{F}(\phi, \theta) = \langle H(\theta) \rangle_\phi - \beta^{-1} S_{\mathrm{vN}}(\rho_\phi)$$

Inner: ϕ tracks thermal state of $H(\theta)$.
Outer: θ reduces KL divergence.

Free-energy minimisation

$\mathcal{F}$ is minimised when $\rho_\phi = \rho(\beta, \theta)$ exactly — the quantum analogue of the EM algorithm.

Free Energy and the QBM Objective

The **quantum free energy** at inverse temperature β:

$$\mathcal{F}(\rho, H) = \mathrm{Tr}[H\rho] - \beta^{-1} S_{\mathrm{vN}}(\rho)$$

$$S_{\mathrm{vN}}(\rho) = -\,\mathrm{Tr}[\rho \ln \rho]$$

The Gibbs state $\rho(\beta) = e^{-\beta H}/Z$ is the *unique* minimiser of $\mathcal{F}$:

$$\rho(\beta) = \arg\min_{\rho} \mathcal{F}(\rho, H)$$

Proof. Using the quantum relative entropy:

$$\mathcal{F}(\rho, H) = \mathcal{F}(\rho(\beta), H) + \beta^{-1}\, S(\rho \,\|\, \rho(\beta)) \geq \mathcal{F}(\rho(\beta), H)$$

since $S(\rho\|\sigma) \geq 0$ (Klein's inequality).

Variational free energy gradient

For a parametrised state ρ_ϕ:

$$\frac{\partial \mathcal{F}}{\partial \phi_k} = \mathrm{Tr}\left[H \, \frac{\partial \rho_\phi}{\partial \phi_k}\right] + \beta^{-1}\, \mathrm{Tr}\left[\ln \rho_\phi \, \frac{\partial \rho_\phi}{\partial \phi_k}\right]$$

The second term involves $\ln \rho_\phi$, which is expensive classically but accessible via Hadamard-test circuits on quantum hardware.

TDVP for imaginary time

The optimal update of ϕ at each step:

$$\sum_j F_{ij}^{(\phi)} \dot{\phi}_j = -\frac{\partial \mathcal{F}}{\partial \phi_i}$$

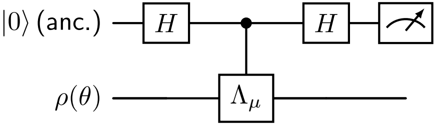
Gradient Computation: Quantum Circuit Approach

Evaluating $\langle \Lambda_\mu \rangle_\theta = \mathrm{Tr}[\Lambda_\mu \, \rho(\theta)]$:

1. Direct measurement (hardware):

1. Prepare $\rho(\theta)$ on the device
2. Measure the Pauli observable Λ_μ
3. Repeat N_{shots} times; estimate error $\sim 1/\sqrt{N_{\text{shots}}}$

2. Hadamard test (for off-diagonal Λ_μ):



$$\langle \Lambda_\mu \rangle_\theta = 2P(\text{ancilla} = 0) - 1$$

3. Parameter-shift for thermal states

For $\Lambda_\mu = P_\mu$ a Pauli operator:

$$\frac{\partial \langle O \rangle_\theta}{\partial \theta_\mu} = \frac{\beta}{2}\left[\langle O \rangle_{\theta_\mu + \delta} - \langle O \rangle_{\theta_\mu - \delta}\right]$$

with $\delta = \pi/(4r)$ where r is the eigenvalue of P_μ: for Paulis $r = 1/2$, so $\delta = \pi/2$.

Important caveat

This thermal parameter-shift rule requires that $\rho(\theta)$ can be re-prepared efficiently for shifted θ — true if the Gibbs state is prepared by a short circuit.

QBM as a Special QNN: Unified View

Both QNNs and QBMs manipulate parameterised quantum states, but the *state type* differs:

Feature	QNN	QBM
State	Pure $\vert\psi(\theta)\rangle$	Mixed $\rho(\theta)$
Objective	Supervised loss	KL divergence
Gradient	Param-shift	Thermal averages
Key tool	QFI / TDVP	Free energy / TDVP
Sampling	Projective	Thermal / Gibbs
Circuits	Unitary	Unitary + trace

Unifying framework

Both are instances of:

$$\min_{\theta} D(\sigma_{\text{target}} \,\|\, \mathcal{E}_\theta(\rho_0))$$

where $\mathcal{E}_\theta$ is a quantum channel (unitary for QNNs, Gibbs map for QBMs) and D is a divergence.

Advantage of QBMs

Mixed-state model $\Rightarrow$ can represent *classically correlated* distributions efficiently, without needing exponentially deep circuits to replicate thermal noise.

Connection to Many-Body Physics

The QBM Hamiltonian lives in the same world as condensed-matter and nuclear physics:

$$H(\theta) = -\sum_{i<j} \theta_{ij}^{zz} Z_i Z_j - \sum_i \theta_i^x X_i - \sum_i \theta_i^z Z_i$$

This is the transverse-field Ising model!

- θ_{ij}^{zz}: spin–spin coupling (Ising)
- θ_i^x: transverse field (quantum fluctuations)
- θ_i^z: longitudinal field (external bias)

Thermal state $\rho(\beta) = e^{-\beta H}/Z$ is the *exact* quantum statistical mechanics object studied in condensed matter — QBM training is parameter estimation for a quantum spin model.

Correspondence table

QBM	Many-body physics
Visible qubits	Measured spins
Hidden qubits	Auxiliary fields
θ_i^x	Transverse field
θ_{ij}^{zz}	Ising coupling
$\rho(\beta)$	Thermal density matrix
Training	Inverse problem

Inverse problem

Given samples from $\rho(\beta, \theta^\star)$, recover $\theta^\star$ — equivalent to learning the couplings of a quantum spin Hamiltonian from measurement data.

QBM and the Variational Principle

The QBM free-energy objective

$$\mathcal{F}(\rho_\phi, H(\theta)) = \mathrm{Tr}[H(\theta)\rho_\phi] - \beta^{-1} S_{\mathrm{vN}}(\rho_\phi) \geq \mathcal{F}_{\mathrm{exact}}$$

provides an **upper bound on the free energy** of $H(\theta)$.

Connection to VQE:

VQE minimises $\langle H \rangle_\psi$ over pure states (zero-temperature limit $\beta \to \infty$). QBM minimises $\mathcal{F}(\rho, H)$ over mixed states (finite temperature).

$$\underbrace{\min_\psi \langle \psi | H | \psi \rangle}_{\text{VQE / ground state}} \quad \subset \quad \underbrace{\min_\rho \mathcal{F}(\rho, H)}_{\text{QBM / thermal state}}$$

TDVP connection

The thermal gradient for ϕ:

$$\sum_j F_{ij}^{(\phi)} \dot{\phi}_j = -\partial_{\phi_i} \mathcal{F}$$

is the *imaginary-time TDVP* — the same equation used in Stochastic Reconfiguration (SR).

QBM training $\equiv$ SR at finite temperature.

$\beta \to \infty$ limit

As $\beta \to \infty$:

$$\rho(\beta) \to |E_0\rangle \langle E_0|$$

Quantum Natural Gradient for QBMs

For the thermal circuit parameters ϕ, the optimal update direction is the **quantum natural gradient**:

$$\Delta\phi = -\eta\, F^{-1}(\phi)\, \nabla_\phi \mathcal{F}$$

where the QFI metric for *mixed states* is:

$$F_{ij}^{(\phi)} = \mathrm{Tr}[\rho_\phi \mathcal{L}_i \mathcal{L}_j]$$

with $\mathcal{L}_i$ the **symmetric logarithmic derivative**:

$$\frac{\partial \rho_\phi}{\partial \phi_i} = \frac{1}{2}(\rho_\phi \mathcal{L}_i + \mathcal{L}_i \rho_\phi)$$

This is the **mixed-state QFI** (Bures metric), which generalises the pure-state QFI from week15.

Bures metric vs. Fubini-Study

State type	Metric
Pure $\lvert\psi\rangle$	Fubini-Study $F_{ij}^{\mathrm{FS}} = 4\,\mathrm{Re}(\ldots)$
Mixed ρ	Bures $F_{ij}^{\mathrm{B}} = \mathrm{Tr}[\rho \mathcal{L}_i \mathcal{L}_j]$

The Bures metric reduces to the Fubini-Study metric for pure states: $F_{ij}^{B} = F_{ij}^{\mathrm{FS}}$ when $\rho = \lvert\psi\rangle\langle\psi\rvert$.

Practical computation

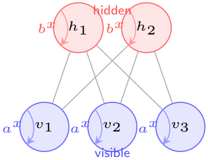
F_{ij}^{B} requires the SLD $\mathcal{L}_i$, which is expensive classically but can be estimated via Hadamard-test circuits

Restricted QBM: Architecture and Hamiltonian

The **Restricted QBM (RQBM)** is the direct quantum analogue of the classical RBM, with n_v visible and n_h hidden qubits:

$$H_{\mathrm{RQBM}} = -\sum_i a_i^z Z_i^v - \sum_j b_j^z Z_j^h - \sum_{ij} W_{ij} Z_i^v Z_j^h - \sum_i a_i^x X_i^v - \sum_j b_j^x X_j^h$$

- a_i^z, b_j^z: classical longitudinal biases
- W_{ij}: inter-layer ZZ coupling (classical part)
- a_i^x, b_j^x: **transverse fields** (quantum)
- No intra-layer ZZ couplings $\Rightarrow$ conditional independence preserved



Classical limit

Set $a_i^x = b_j^x = 0$: recovers exact classical RBM.

Self-loops denote transverse fields a_i^x, b_j^x (quantum fluctuations within each unit).

RQBM Gradient and Contrastive Divergence

For the RQBM, the gradient of the log-likelihood w.r.t. the inter-layer coupling W_{ij} is:

$$\frac{\partial \ln p(\mathbf{v})}{\partial W_{ij}} = \beta \left[\langle Z_i^v Z_j^h \rangle_{\text{clamp}} - \langle Z_i^v Z_j^h \rangle_{\text{free}} \right]$$

For the transverse-field bias a_i^x:

$$\frac{\partial \ln p(\mathbf{v})}{\partial a_i^x} = \beta \left[\langle X_i^v \rangle_{\text{clamp}} - \langle X_i^v \rangle_{\text{free}} \right]$$

Key difference from classical: the quantum gradient involves $\langle X_i \rangle$ — an off-diagonal expectation value with no classical analogue.

Quantum CD algorithm

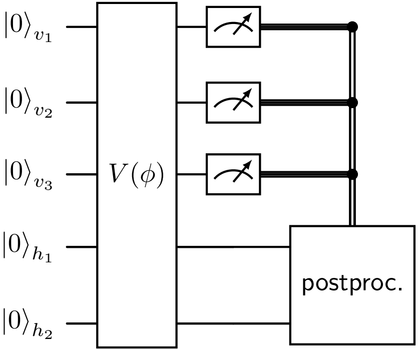
1. **Positive phase:** clamp v to data; sample h from $\rho_h^{(v)} = \mathrm{Tr}_v[\rho_{\text{clamp}}]$
2. **Negative phase:** run k steps of quantum Gibbs sampling (Metropolis or QITE) to approximate ρ_{free}
3. **Update:** $\Delta W_{ij} \propto \langle Z_i Z_j \rangle_+ - \langle Z_i Z_j \rangle_-$

Advantage over classical CD

Quantum sampling of $\langle X_i \rangle$ and $\langle Z_i Z_j \rangle$ is naturally unbiased on quantum hardware — no MCMC convergence issue.

RQBM Circuit: Sampling and Inference

Sampling from the RQBM: prepare the Gibbs state and measure the visible qubits.



Training (positive phase): clamp visible registers to data sample $\mathbf{v}$ and measure $\langle Z_i^v Z_j^h \rangle$, $\langle X_i^v \rangle$, $\langle X_j^h \rangle$.

Training (negative phase): run $V(\phi)$ freely; measure the same observables to estimate the model expectation values $\langle Z_i Z_j \rangle_{\text{free}}$.

Expressivity of QBMs vs. Classical BMs

What a classical RBM can represent

- Any probability distribution over $\{0,1\}^n$ with sufficiently many hidden units
- But: requires exponentially many hidden units to replicate some quantum distributions
- Limited to distributions arising from diagonal (classical) Hamiltonians

Formal expressivity result:
Any n-qubit density matrix ρ can be expressed as the Gibbs state of some n-qubit Hamiltonian:

$$\rho = \frac{e^{-\beta H}}{Z}, \quad H = -\beta^{-1} \ln(Z\rho)$$

Key consequence: the QBM is a *universal* model for quantum states.

What a QBM adds

- Off-diagonal density matrix elements $\Rightarrow$ coherences
- Entangled marginals over visible qubits
- *Quantum-correlated* probability distributions — not realisable by any classical latent variable

Sample complexity

Learning ρ to within ε in trace distance requires:

$$N = O\left(\frac{n}{\varepsilon^2}\right) \text{ samples}$$

(quantum state tomography lower bound)

When Do QBMs Outperform Classical BMs?

Theoretical separation:

1. **Quantum data** (molecular wavefunctions, lattice gauge configs): target is inherently quantum; a QBM learns it directly.
2. **Long-range entanglement:** exponentially many classical hidden units needed; $O(n)$ quantum hidden qubits suffice.
3. **Partition function estimation:** classical Monte Carlo suffers sign problems for fermionic systems; QBM avoids this.

Practical regime (NISQ):

- Small n ($\lesssim 30$ qubits)
- Shallow thermal circuits (short imaginary time)
- Heuristic advantage — not yet provable

Caution: no-go results

For *classical* data distributions:

- Any QBM can be dequantised (matched by a classical BM) if the quantum correlations of ρ are weak.
- Advantage requires ρ to have genuinely quantum off-diagonal structure.

Barren plateaus in QBMs

Like QNNs, deep QBM circuits suffer exponentially vanishing gradients:

$$\mathrm{Var}(\nabla_\theta \mathcal{L}) \sim 2^{-n}$$

Mitigation: shallow circuits and problem-inspired Hamiltonian structure

NumPy Simulation of a Small QBM

```python
import numpy as np
from scipy.linalg import expm

def gibbs_state(H, beta=1.0):
    """Exact␣Gibbs␣state␣rho␣=␣exp(-beta*H)␣/␣Tr[exp(-beta*H)]."""
    evals, evecs = np.linalg.eigh(H)
    weights = np.exp(-beta * evals)
    weights /= weights.sum()
    rho = sum(w * np.outer(v, v.conj())
              for w, v in zip(weights, evecs.T))
    return rho

def qbm_hamiltonian(theta_z, theta_x, W, n_v, n_h):
    """Transverse-field␣Ising␣RBM␣Hamiltonian."""
    n = n_v + n_h
    I2 = np.eye(2); Z = np.diag([1., -1.]); X = np.array([[0.,1.],[1.,0.]])
    def kron_op(gate, i):
        ops = [gate if j == i else I2 for j in range(n)]
        M = ops[0]
        for op in ops[1:]: M = np.kron(M, op)
        return M
    H = np.zeros((2**n, 2**n), dtype=complex)
    for i in range(n):      H -= theta_z[i] * kron_op(Z, i)
    for i in range(n):      H -= theta_x[i] * kron_op(X, i)
    for i in range(n_v):
        for j in range(n_h):
```

QBM Training Loop (NumPy)

```
def qbm_gradient(theta_z, theta_x, W, data, n_v, n_h, beta=1.0):
    """Estimate gradient of negative log-likelihood."""
    n = n_v + n_h
    I2 = np.eye(2); Z = np.diag([1., -1.])
    H = qbm_hamiltonian(theta_z, theta_x, W, n_v, n_h)
    rho_free = gibbs_state(H, beta)

    dW = np.zeros_like(W)
    for v_sample in data:
        # Clamp visible qubits to data sample
        H_clamp = clamp_hamiltonian(H, v_sample, n_v)
        rho_clamp = gibbs_state(H_clamp, beta)
        for i in range(n_v):
            for j in range(n_h):
                ZZ = kron_op(Z, i, n) @ kron_op(Z, n_v+j, n)
                pos = np.real(np.trace(ZZ @ rho_clamp))
                neg = np.real(np.trace(ZZ @ rho_free))
                dW[i, j] += beta * (pos - neg)
    return dW / len(data)

# Training loop
lr = 0.05
for step in range(200):
    grad_W = qbm_gradient(theta_z, theta_x, W, data, n_v, n_h)
    W += lr * grad_W
    if step % 50 == 0:
```

Exercises

1. **Classical limit.** Set $\theta_i^x = 0$ for all i in the RQBM. Show that the Gibbs state becomes diagonal in the Z-basis and the gradient reduces to the classical CD update.
2. **Free energy minimisation.** For a single qubit with $H = -\theta^x X - \theta^z Z$, compute the exact Gibbs state $\rho(\beta)$, the free energy $\mathcal{F}(\rho, H)$, and verify that $\rho(\beta)$ minimises $\mathcal{F}$ over all single-qubit density matrices.
3. **TFD circuit.** Implement the TFD preparation circuit for $n = 2$ qubits using NumPy. Verify that tracing out the ancilla reproduces the Gibbs state up to circuit approximation error.
4. **Bures vs. Fubini-Study.** For a pure state $\rho = |\psi\rangle\langle\psi|$ show analytically that $F_{ij}^B = \mathrm{Tr}[\rho\mathcal{L}_i\mathcal{L}_j]$ reduces to $F_{ij}^{\mathrm{FS}} = 4\,\mathrm{Re}(\langle\partial_i\psi|\partial_j\psi\rangle - \langle\partial_i\psi|\psi\rangle\langle\psi|\partial_j\psi\rangle)$.
5. **Quantum advantage.** Construct a 2-qubit density matrix ρ with off-diagonal coherences not representable by any classical BM. What is the minimum number of hidden qubits needed?

Outlook: Open Problems and Future Directions

Near-term (NISQ) focus

- Small QBMs ($n \lesssim 30$) on current hardware
- Hybrid: quantum sampling + classical update
- Applications: quantum chemistry, finance, combinatorial optimisation
- Error mitigation for thermal state preparation

Key open questions

- Can QBMs learn classically hard-to-sample distributions efficiently?
- Can barren plateaus be avoided with problem-inspired structure?
- What is the quantum sample complexity advantage over classical BMs?

Long-term directions

- **Fault-tolerant QBMs:** exact Gibbs state prep via quantum signal processing
- **Quantum GANs:** replace discriminator or generator with a QBM
- **Physics:** learn Hamiltonians from tomographic data; phase diagram estimation; sign-problem-free MC via QBM proposals
- **Tensor networks:** finite-T DMRG $\leftrightarrow$ shallow QBM circuits

Quantum advantage landscape:

$$\underbrace{\text{Classical BM}}_{\text{classical data}} \subset \underbrace{\text{QBM}}_{\text{quantum data}} \subset \underbrace{\text{Full QC}}_{\text{fault-tolerant}}$$

Summary: QBM in Context

	Classical BM	**QNN**	**QBM**
State	Distribution $p(\mathbf{v})$	Pure $\lvert\psi(\theta)\rangle$	Mixed $\rho(\theta)$
Model	$e^{-\beta E}/Z$	$U(\theta)\lvert 0\rangle$	$e^{-\beta H(\theta)}/Z$
Gradient	CD (biased)	Param-shift (exact)	Thermal avg. (exact on QC)
Geometry	Euclidean	Fubini-Study QFI	Bures metric
Training	Gibbs sampling	TDVP (real time)	TDVP (imag. time)
Expressivity	Classical dists.	Pure states	All quantum states
Advantage	Tractable	Discriminative	Generative

The central message

A Quantum Boltzmann Machine is a **thermal QNN**: it replaces the pure-state variational ansatz with a Gibbs (mixed) state, enabling universal generative modelling of quantum distributions with a natural connection to statistical mechanics and the variational principle.

References

Quantum Boltzmann Machines

1. M. H. Amin et al., *Quantum Boltzmann machine*, Phys. Rev. X **8**, 021050 (2018).
2. G. Verdon et al., *A quantum algorithm to train neural networks using low-depth circuits*, arXiv:1712.05304 (2017).
3. J. Gao & L.-M. Duan, *Efficient representation of quantum many-body states with deep neural networks*, Nat. Commun. **8**, 662 (2017).
4. C. Zoufal et al., *Variational quantum Boltzmann machines*, Quantum Mach. Intell. **3**, 7 (2021).

Thermofield double and Gibbs state preparation

5. S. Chowdhury et al., *A variational quantum algorithm for preparing quantum Gibbs states*, arXiv:2002.00055 (2020).
6. D. Zhu et al., *Generation of thermofield double states and critical ground states with a quantum computer*, PNAS **117**, 25402 (2020).

QNN and QFI (see also week15)

7. J. Stokes et al., *Quantum natural gradient*, Quantum **4**, 269 (2020).
8. M. H. Amin, *Searching for quantum speedup in quasistatic quantum annealers*, Phys. Rev. A **92**, 052323 (2015).
9. R. Babbush et al., *Focus beyond quadratic speedups for error-corrected quantum advantage*, PRX Quantum **2**, 010103 (2021).